

MUHAMMAD HISYAM

Full Stack & AI Software Engineer

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PROFESSIONAL SUMMARY

Full Stack & AI Software Engineer with extensive expertise in building desktop cross-platform applications, high-performance web systems, medical deep learning pipelines, and agentic AI integrations. Skilled across TypeScript, Rust, Python, Golang, Java, and Supabase/PostgreSQL. Creator of 26+ open-source and commercial-grade projects including RetailHub (Tauri + Supabase POS platform), Pneumonia U-Net Segmentation, and Gemini MCP Client.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Rust, Python, Golang, Java, PHP, PL/pgSQL, HTML5, CSS3, LaTeX

Desktop & Frontend: Tauri, Astro 5, React.js, SolidJS, SvelteKit, Next.js, Vue.js, Tailwind CSS, TanStack Router

Backend & Cloud: Supabase, Node.js, Express.js, Go (Gin/Fiber), Java Spring Boot, Laravel, REST APIs, WebSockets

AI & Data Engineering: PyTorch, U-Net, OpenCV, Model Context Protocol (MCP), Gemini API

Databases & DevOps: PostgreSQL, Supabase RLS, MongoDB, Redis, Docker, Git, GitHub Actions, Vite, Bun, Biome, Linux CLI

FEATURED PROJECTS

RetailHub – Retail Management & POS Platform (Tauri + Rust + Supabase) [GitHub Repository](#)

- Engineered a desktop and cross-platform Retail Point-of-Sale (POS) and inventory management platform using Tauri (Rust backend runtime), TypeScript, Vite, and TanStack Router.
- Configured Supabase (PostgreSQL) data architecture with Row Level Security (RLS) policies, PL/pgSQL database functions, and real-time database subscription sync.

Developer Portfolio & Interactive Hub (Astro 5 + SolidJS) [GitHub Repository](#)

- Engineered a high-speed developer portfolio using Astro 5, SolidJS, and Tailwind CSS 4.
- Integrated an embedded Dev Suite featuring a retro Unix terminal shell (`hisyam-cli`), REST API response demo client, audio music player, and 8 mini-games arcade.

Pneumonia Segmentation U-Net Model (Python + Deep Learning) [GitHub Repository](#)

- Developed a deep learning semantic segmentation pipeline utilizing PyTorch and U-Net architecture for chest X-ray pneumonia detection.
- Implemented custom loss metrics, data augmentation transformers, and inference evaluation routines.

Gemini Model Context Protocol (MCP) Client (TypeScript) [GitHub Repository](#)

- Built an extensible MCP client enabling agentic AI workflows and tool-use integrations with Google Gemini models.

Real-Time Energy Monitoring System (TypeScript + IoT) [GitHub Repository](#)

- Built a real-time energy telemetry analytics dashboard processing high-frequency sensor streams with dynamic live charts.

Golang & SvelteKit Real-Time Chat App (Go + WebSockets)

[GitHub Repository](#)

- Implemented a concurrent real-time chat application with a Golang WebSocket backend server and SvelteKit reactive frontend.

Secure JWT Authentication with Refresh & Version Tokens (Java)

[GitHub Repository](#)

- Designed a robust Java authentication microservice supporting access tokens, sliding session refresh tokens, and instant token version invalidation.

EDUCATION & COMMUNITY

Bachelor of Science in Computer Science / Information Technology

Indonesia

Focus on Software Engineering, Web Systems, and Applied AI

- Maintainer of 26+ open-source GitHub repositories spanning AI, web apps, backend APIs, and developer utilities.
- Contributor to collaborative software projects including SMKN 32 SmartAssist AI Chatbot and Woven web platform.